

Total No. of Questions : 4]

SEAT No. :

PD263

[Total No. of Pages : 1

[6411]-38

B.E. (Civil Engineering) (Insem)

GREEN STRUCTURES & SMART CITIES

(2019 Pattern) (Semester-VIII) (401014 E) (Elective-VI)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat sketches must draw wherever necessary.
- 3) Figures to right indicate full marks.

- Q1)** a) Define embodied energy and explain guidelines to reduce embodied energy. [5]
- b) Define ecological footprint and explain method of estimating ecological footprint. [5]
- c) Enlist criteria for selection of green building materials and explain Indoor Air quality. [5]

- Q2)** a) Explain Sustainable Green Building Materials. [5]
- b) Deliberate the difference between Embodied energy and life cycle energy. [4]
- c) Explain bio capacity and deliberate the causes of environment problems of ecological footprint? [5]

- Q3)** a) What is Sustainable sitting and explain principles of green construction practices. [5]
- b) Enlist green rating systems and describe any two of them. [5]
- c) Explain components of water balance. [5]

OR

- Q4)** a) Explain net zero energy building. [5]
- b) Explain effect of trees and microclimatic modification through greening. [5]
- c) Explain operational energy reduction in detail. [5]

